phosphates in the root zone of plants and satisfy the phosphorus nutrition of plants.

Morphology

VAM is an example of endomycorrhiza meaning, the storage organelles of phosphates like vesicles and arbuscles are seen intracellularly. Vesicle is a globose structure and arbuscle is a tree like branching structure present in the root cortical cells (Figure 11.9). VAM fungi are naturally most prevalent in angiosperms. gymnosperms, pteridophytes and bryophytes.



Figure 11.9: The fresh water fern *Azolla.*

Mass production

Root based inoculum is used for preparing VAM biofertilizer (Figure 11.10). The selected spores of VAM fungi are allowed to infect plants like onion, sorghum and other grasses. After 3-4 months, the roots along with the soil are macerated or pelleted with an inert material and packed in polythene pouches which can be used as biofertilizer.

11.2.4 Cyanobacteria / Blue green Algae

Blue green algae are single celled or filamentous prokaryote capable of nitrogen

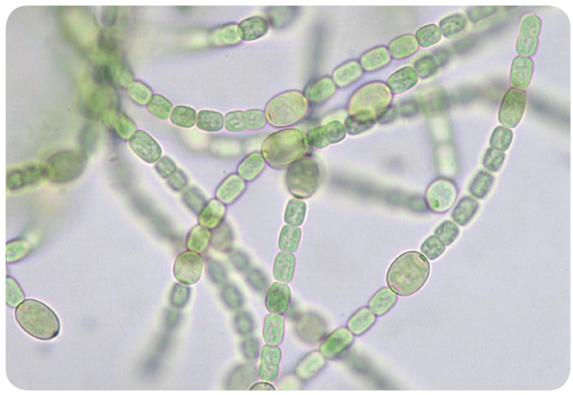


Figure 11.10: Microscopic view of *Anabaena azollae*

fixation and photosynthesis. Most of the filamentous forms have specialized large, thick walled cells called heterocysts which are sites of nitrogen fixation.

Example: *Nostoc*, *Anabaena* is examples for filamentous BGA. *Gleocapsa* is an example of unicellular BGA. Some of the filamentous forms do not possess heterocysts but still fix atmospheric nitrogen. Since they need standing water for their growth, BGA can effectively colonize paddy fields and enrich the soil with nitrogen.

Mass cultivation of BGA

Applying BGA to paddy fields can reduce the amount of chemical nitrogenous fertilizer applied for the growth of paddy crop. Therefore cultivation of BGA in large quantities is necessary. Mass cultivation of BGA has the following steps.

1. Isolation of BGA
2. Mass cultivation of BGA

Isolation of BGA

BGA can be isolated from soil or paddy fields. Appropriate dilutions from serially diluted algal sample are inoculated in the liquid flasks containing algal media

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